

Quiz 1 review.

Proof. $\sqrt{3}$ is irrational

We will need the following lemma:

- ① Lemma: For $\forall a, b \in \mathbb{Z}$, if a and b are odd integers, then their product ab is also odd
- ② Lemma: Any rational $x: \frac{m}{n}$ can be written as a ratio of two integers m, n without common factors
- ③ Lemma: Even plus even is even
Odd plus odd is even
Odd plus even is odd
Even plus odd is odd

From Lemma ①. $p \rightarrow q$

So we can say $\neg p \rightarrow \neg Q$

that is : For $\forall a, b \in \mathbb{Z}$, if a or b are even integers, then their product ab is even

We assume that $\sqrt{3}$ is a rational

From Lemma 2, we can represent

$\sqrt{3}$ as $\frac{m}{n}$, where m, n don't have common factors and $n \neq 0$

We have : $(\sqrt{3})^2 = 3 = \frac{m^2}{n^2}$

$$\frac{m^2}{n^2} - 1 = \frac{m^2 - n^2}{n^2} = 3 - 1 = 2$$

$$(m+n)(m-n) = 2n^2$$

If m, n are both even :

then m, n have a common factor 2 (X)

If one of m, n is even and the other one is odd :

From Lemma ② : $2n^2$ is even

From Lemma ③ : $(m+n)$ and $(m-n)$
are odd

From Lemma ② : $(m+n) \cdot (m-n)$ ~~(X)~~
 $\Rightarrow \text{odd} \neq \text{even} \Rightarrow 2n^2$

If m, n are both odd.

From Lemma ② : $2n^2$ is even

From Lemma ③ : $(m+n), (m-n)$ are
both even

$$(m+n), (m-n) \neq 0$$

Assume that $m+n = 2i$, $m-n = 2j$, $i, j \in \mathbb{Z}$

We have : $2i \cdot 2j = 2n^2$

$$2i \cdot j = n^2$$

From Lemma ② : $2i \cdot j \Rightarrow \text{even}$

$$n^2 \Rightarrow \text{odd}$$

$\text{even} \neq \text{odd}$ ~~(X)~~

Therefore, we see that $\sqrt{3}$ is not a rational

It is a irrational